



2A

How did space exploration begin?

ENGLISH

- Why do we explore space?
- Why is exploring space difficult? ?

LISTENING

1 Look at each picture. Then look at the sentence. With a partner, discuss the questions.

space: everything above the Earth.
lunar: the Moon



The space rocket launches towards space.

- What is a rocket?
- Why is it called a **space** rocket?
- What does the verb **launch** mean?
- Where is the rocket launching to?
- Why is the rocket going there?

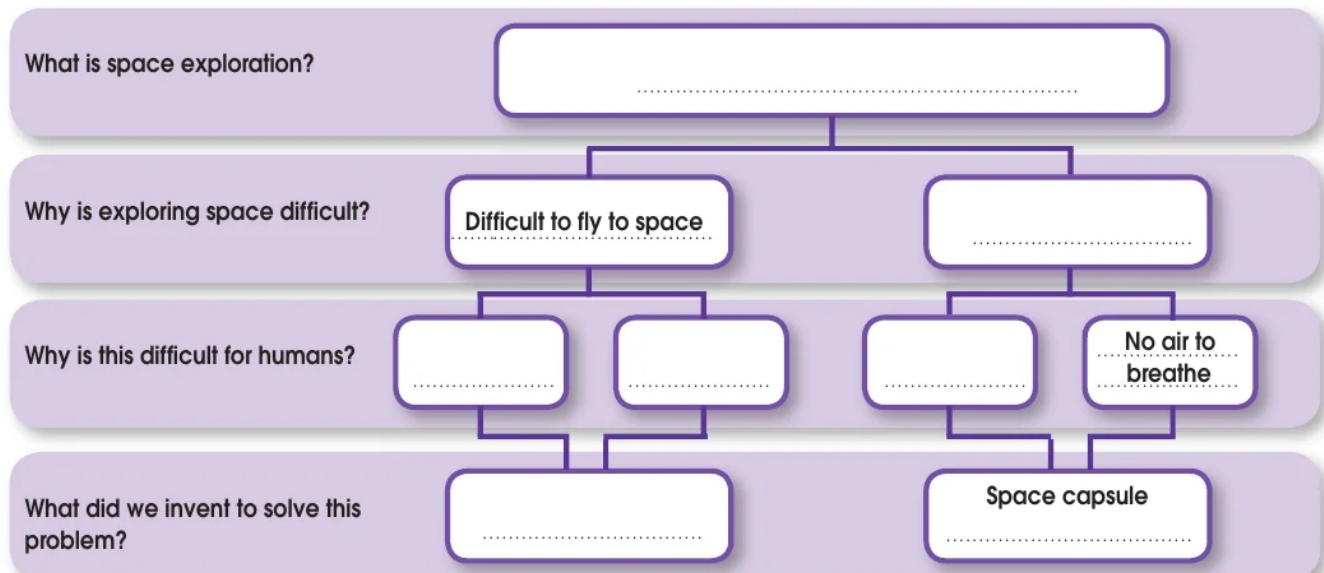


The lunar capsule lands on the Moon.

- What is a capsule?
- Why is it called a **lunar** capsule?
- What does the verb **land** mean?
- What is the lunar capsule landing on?
- Why is the capsule going there?

LISTENING

2 Look at the mind map. What is space exploration? Write your definition in the first box.
You are going to listen to a radio documentary about space exploration. Listen to the first part. Fill in the missing information in the mind map.

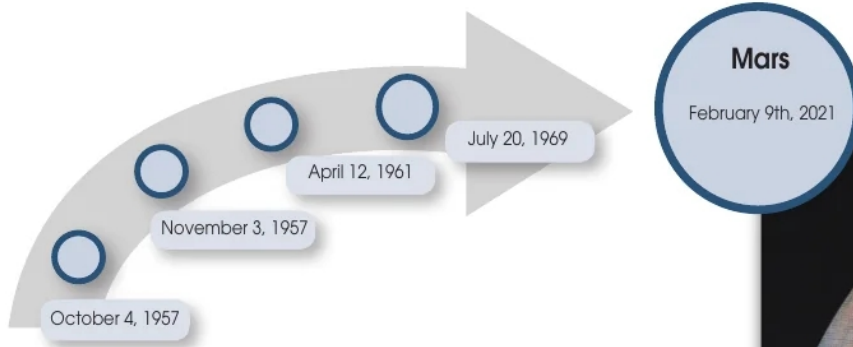


3  Listen to the second part of the documentary. Match the event to its date on the timeline.

- 1 Laika the dog is the first animal to travel in a space capsule.
- 2 The astronaut Neil Armstrong is the first man on the moon.
- 3 The first satellite is launched into space.
- 4 The astronaut Yuri Gagarin is the first man in space.

Did You Know?

To reach space, a space rocket needs a speed of at least 28,440 km/h. The fastest car in the world only goes 490 km/h.



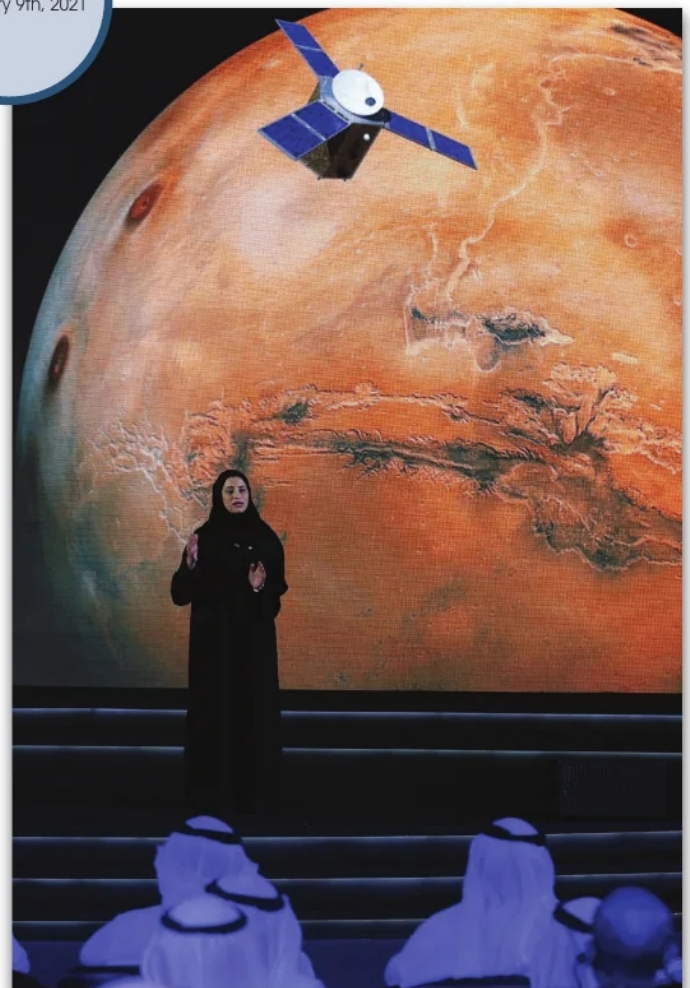
C

In your group, you will be either the presenter or a journalist asking questions.

Each journalist will choose one question from the list below. The presenter needs to prepare an answer for each question.

Look at the audio script at the back of the coursebook to help you prepare your answers.

Why did it take humans so long to explore space?
 Can humans live in space?
 Why did they send animals into space first?
 What does a satellite do?
 What do humans need to explore the Moon?
 What is the next step in space exploration?





2A

How did space exploration begin?

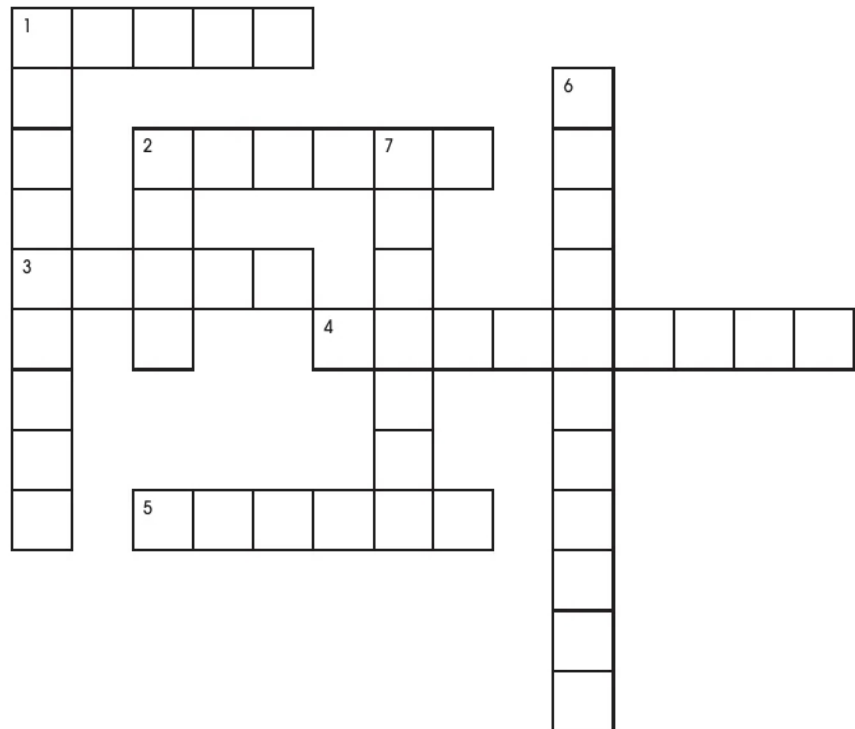
- 1** Read the definitions. Write the vocabulary word that matches the definition. The first letter of the word has been provided.

Across

- 1 Everything beyond the Earth. (S-)
- 2 To send something in the air. (L-)
- 3 Describes the Moon. (L-)
- 4 Someone who explores space. (A-)
- 5 A long spacecraft that can travel quickly. (R-)

Down

- 1 An object sent into space to collect information. (S-)
- 2 To bring something down to the ground. (L-)
- 6 The activity of going somewhere never visited by humans. (E-)
- 7 A small container where astronauts live. (C-)



A space rocket launching from a **launch pad**.

Language Tip

Compound nouns are two words put together to make a new word. The first word in a compound noun usually describes the second word. For example:

Launch means to send something in the air.

Pad is a flat surface.

A *launch pad* is a flat surface where rockets are launched into space.

- 2** Look at the definitions for the vocabulary words in Activity 1. Use the definitions to help you define the compound words below.

- 1 Spacesuit:
Astronauts wear this suit to protect them in space.
- 2 Lunar landing:
- 3 Space capsule:
- 4 Space rocket:
- 5 Lunar exploration:

3 Match the words below to make a new word. Then, match the new word to its definition.

- | | | | |
|---|-------|---------|---|
| 1 | lunar | light | the air astronauts breathe in the capsule |
| 2 | air | landing | to fly through space |
| 3 | sun | supply | to land a capsule on the Moon |
| 4 | up | flight | to move up |
| 5 | space | ward | the light from the sun |

4 Look at the four important events on the timeline in Activity 3 in the Coursebook. You will make your own timeline of important events in your life. Think of four important events in your life. Write down the date when the events happened. Then write down what those events were.

- Date
- Event 4



- Date
- Event 3



- Date
- Event 2



- Date
- Event 1



2B

What would you put in a time capsule?

ENGLISH

- What is a time capsule?
- Why do people make them?

LISTENING

- 1 In 1977, scientists sent an audio/visual time capsule into space called the **Golden Record**. What do you think was on it? Listen to the first part of the recording and put a check next to the items mentioned.

animal sounds • a picture of New York
 recipes for cooking • a greeting in Arabic
 stories • images of famous paintings

- 2 Listen to the second part of the recording. Answer the questions.

- 1 What is Sultan's homework?
- 2 Is he going to use a record? If not, what is he going to use?
- 3 What place does Sultan say he would include? Does Abdullah agree? Does Nawaf agree?
- 4 Does Sultan agree with Nawaf's idea?

- 3 Listen to the second part of the recording again. Fill in the missing words.

- ... my homework is to decide what
 I if
 I a time capsule to send to space.
- ... but mostly, I
 information about the UAE.
- I Burj Khalifa,
 because it's the tallest building in the world.
- I a bunch of
 photos and videos.

- 4 Listen to sentences from the recording. Do you hear **would** or **'d**?

1would..... 2 3 4 5 6



SPEAKING

- 5 Think of what to include in your own space time capsule

A video going up Burj Khalifa

- 6 Compare your answers in groups.
If I made a space time capsule,

I would include
because

I would include
because

- 7  In your groups, decide on six items to include in your space time capsule.

A video going up Burj Khalifa

SPEAKING TIP

You can use these phrases to agree or disagree with someone.

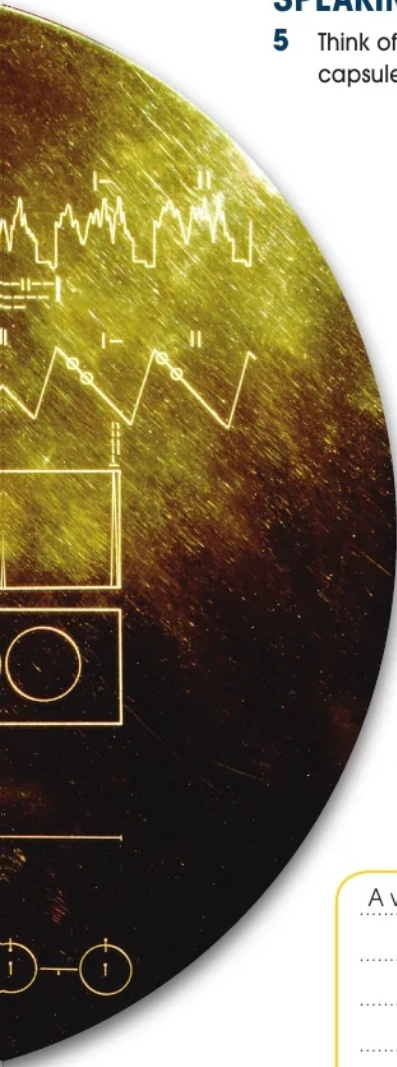
You're right.

Me too.

I'm not so sure.

I agree.

I don't really agree.





2B

What would you put in a time capsule?

1 Match the words to their definitions.

- | | |
|-----------------|---|
| 1 an inch | a a person who travels to space |
| 2 clip | b to look for information |
| 3 thunderstorms | c made of gold; gold-colored |
| 4 astronaut | d 2.54 centimeters |
| 5 disc | e a person who works in science |
| 6 research | f a round, flat item, like a DVD, CD, or record |
| 7 scientist | g a short audio or video file |
| 8 golden | h bad weather with rain, thunder, and lightning |

2 Read the facts about the **Golden Record**. Fill in the blanks with the words from Activity 1.

The Golden Record

In 1977, after years of ¹ research, ² _____ from NASA sent the ³ _____ Record to space on Voyager 1. This golden ⁴ _____ was sent with no ⁵ _____, because humans are still unable to travel great distances in space. The twelve- ⁶ _____ disc is a sort of audio/visual time capsule that contains information about our world. There are greetings in 55 languages, sound ⁷ _____ from Earth, such as wind, rain, and ⁸ _____ – and even animal noises. There are even pictures of important places, like the Taj Mahal or Sydney Opera House. The Golden Record has been in space for over forty years now. Even after it stops sending messages to Earth, it will continue traveling at almost 50,000 km/h through space.

3 Write three sentences using some of the words from Activities 1 and 2.

- 1
- 2
- 3

Language Tip

Read the sentence:

*If I made a space time capsule,
I would put photos of the Burj
Khalifa and Sheikh Zayed Grand
Mosque in it.*

This is an example of the **Second Conditional**. The **Second Conditional** uses the *past simple* and *would*, but talks about now or the future.

4 Look at the phrases in the box. Decide which are for agreeing and which for disagreeing..

You're right. I'm not so sure. Me too. I agree.

Agreeing	Disagreeing

5 Match the sentence halves together..

- | | |
|--|--|
| 1 If NASA sent out more Voyagers, | a if another space time capsule were made. |
| 2 It would be better to use a tablet or computer | b what would you include? |
| 3 If you made a space time capsule, | c if they made another Golden Record. |
| 4 If I made a space time capsule, | d they would take many years to travel to the edge of the solar system |
| 5 Nasa would probably send the same information | e I would include pictures of Burj Khalifa. |

6 Complete the dialogue. Use phrases from the Speaking Tip and 'd where possible.

Jamila: What ¹ would you put on your tablet, or, time capsule?

Wadeema: I'm not sure – it's a difficult decision.

Noura: ².....There are so many things to choose from!

Wadeema: I ³..... include the Emirates Palace, because it's such an amazing building. I ⁴.....put a bunch of photos and videos, especially ones taken from the beach

Noura: ⁵..... And don't forget the Burj Al Arab. ⁶.....also include lots of photos of its brilliant architecture.

Jamila: ⁷..... ⁸.....have to have more than buildings. What about a video of someone skydiving over the Palm Jumeirah?

Wadeema: ⁹..... – I think we should put all of it in the time capsule!



What would you take on a space journey?

ENGLISH

- What do you think living in space is like?
- Would you like to go to space?

READING

- 1** Read the magazine interview quickly. Who is Abdulla Al Shamsi? What does he talk about in the interview? Circle the correct answers from the list below:

- basketball
- rockets
- space travel
- living on Mars

- 2** Read the interview again. Which one word can fit in each space?

- 3** Read again and answer the questions.

- 1 Is Abdulla going to space in the future? **Yes** or **No**
- 2 Did Abdulla go to space in the past? **Yes** or **No**
- 3 Would Abdulla go to space if he had the chance? **Yes** or **No**
- 4 What would Abdulla take to space?
.....
- 5 What would Abdulla take to Mars?
.....

Celebrity Interview!

Abdulla Al Shamsi is a famous athlete who has written a science-fiction book about living on Mars. **Abdulla, why did you write this book?**

"I wanted to make my family proud. I like science fiction and I'm interested in space exploration, so one day I just started writing about it."

Do you think humans will travel to Mars in the next 10 years?

"It's possible. I think we will go when we have the best technology and the best people. The astronauts have to be strong because they might be living on Mars for a very long time. They can't just come back."

If you were an astronaut, would you go to Mars?

"Of course I It be an amazing experience."

If I could take my family, it be even better."

If you went to space, what would you take?

"If I went to space, I take my camera."

The photos be amazing! Sports be difficult because the gravity is different. So, I n't take my tennis racket."

If I went to Mars, I take food and seeds. I imagine I be there for many years. I grow plants and vegetables on my ship and on Mars. I also take a notebook and I write more books."

Do you think you will write another book?

"Yes, if I have time, I will write another science fiction book next year."



SPEAKING

- 4 21 If you went to space, what would you take?



LANGUAGE FOCUS

- 5 Read the two sentences below. How are the meanings different?

- 1 If I go to space, I will be happy.
If I went to space, I would be happy.

.....

- 2 Look at the text on page 37 again and find the **negative** sentence.
How do you make a negative second conditional sentence?

.....

- 3 Look at the text again and circle the questions.
How do you make a **question** using the second conditional?

.....

.....

Use of English

We use the **second conditional** to talk about something we imagine. It can be in the present or the future, but it is not real. Look at the examples.

If Omar had a plane, he would travel to many countries.

Does Omar have a plane?

Is he planning to get a

plane?

If + past simple, I would + bare infinitive verb.

1

2

If I **went** to space, I **would take** a camera.

I **would** take a camera if I **went** to space.



SPEAKING

- 6 Imagine you could interview a famous person. Who would you interview? What questions would you ask? Talk to a partner and share your ideas..



2C

What would you take on a space journey?

1 Match the words with their definitions.

- | | |
|-------------------|--|
| 1 to imagine | a someone who travels to space |
| 2 astronaut | b plants grow from these |
| 3 seeds | c to think or dream |
| 4 science fiction | d a type of movie or book about space |
| 5 to be proud | e to feel happy because you or someone you know did something good |

2 Put the words in the correct order to make sentences.

- grow vegetables / if / went / to space / I / I / would
If I went to space, I would grow vegetables.
- I / to space / would / went / if / I / my camera / take
.....
- you / would / go / to Mars / ?
.....
- you / would / if / had / a rocket / where / go / ? / you
.....
- wouldn't / to the sun / go / I
.....
- be excited / a trip to space / would / won / I / if / I
.....

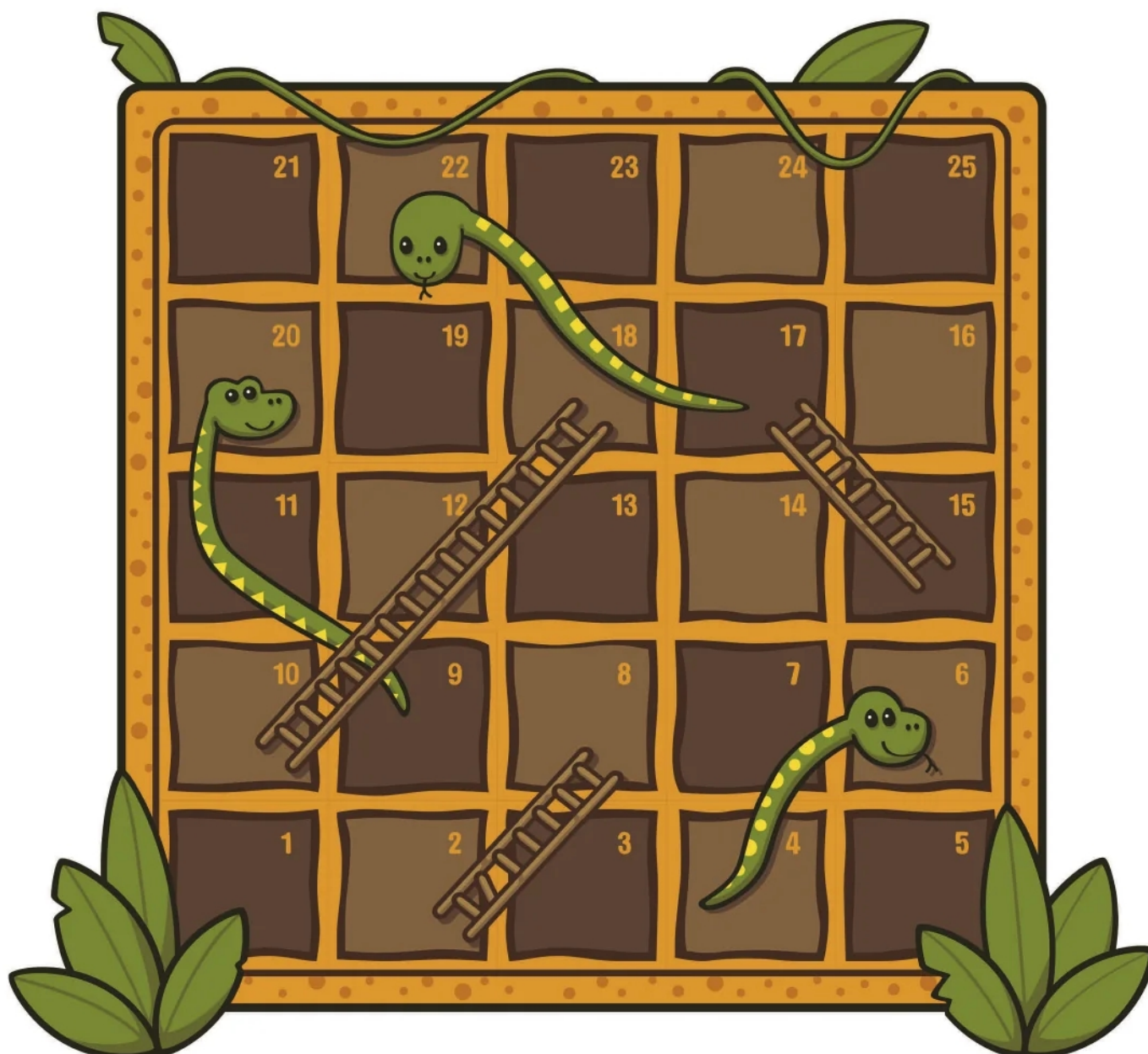
3 Who would you interview if you had the chance? What questions would you ask? Write your questions below and then role play your interview with a partner.

An interview with

- Questions:
-
 -
 -
 -

4 Space Snakes Board Game

- Get a counter.
- Start on Square 1
- Flip a coin:
- If you land on an *even number*, make a second conditional sentence.
2 4 6 8 10 12 14 16 18 20 22 24
- If you land on an *odd number*, choose a friend and ask a second conditional question.
1 3 5 7 9 11 13 15 17 19 21 23 25
- If you land on a snake, go back down!
- If you land on a ladder, climb up!
- The winner is the first person to get to square 25!





What does an astronaut do?

- What do you know about the International Space Station?
- What do you think scientists do at the International Space Station?

READING

1 Circle the best title for the article:

- 1 Daily Life of an Astronaut
- 2 Growing Vegetables in Space
- 3 A Flight to the Moon!

A. The UAE Space Newsletter has an exciting story from our very own astronaut, Mona, who is currently stationed at the International Space Station (ISS) in outer space.

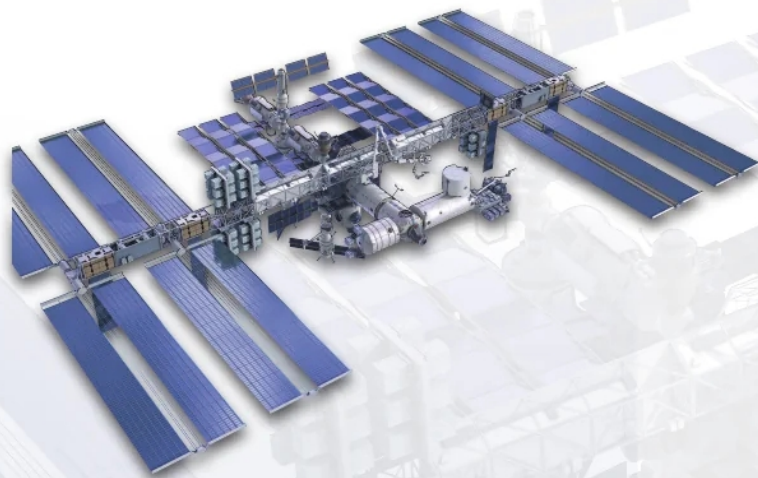
B. Today was my 8th day at the space station. It has been an exciting trip. My mission is to help conduct science **experiments** to **grow vegetables**, like **zucchini**, in space. This will help future astronauts learn how to grow food on long space missions. So, I have brought many different plant seeds with me.

To get to the ISS, we had to get into a big **rocket**. We were flying over 28,000 kilometers per hour. It is a two-day trip to the ISS. When we arrived, we were greeted by the ISS **crew**. They were **floating** in the air because they were in zero **gravity**.

C. Now that we have been at the ISS for over a week, I have started my experiments. I am testing how much light and water is needed to grow different plants in space. This is important because long space flights in the future will have a small amount of water and **oxygen** to grow plants. They will also have a small amount of energy from batteries to use on light. So, it is important that we understand which plant is the most efficient to grow on long space missions.

Reading Strategy

Read for gist by reading the text quickly to understand the general idea. After each paragraph, think of one or two words that can summarize the whole paragraph.



I planted three plants in three separate pots under the same growing conditions. The first pot is growing zucchinis. The second pot is growing potatoes. The third pot is growing corn. Each plant will receive six hours of sunlight and four centimeters of water. In three weeks, I will write about the results of the experiment ...

- 2 Read the text again. How much sunlight and water will Mona give each plant?
- 3 Read the text again and then look at the table below that shows the different growing conditions for each vegetable. In pairs, predict what the results of the experiment will be. Give reasons.

	Water (centimeters weekly)	Sunlight (hours daily)
Zucchini	3 cm	8-10
Potatoes	2.5-5 cm	6
Corn	5-6 cm	7.5

- 1 Which vegetable will grow the best?

.....

.....

.....

- 2 Which vegetable will grow the least?

.....

.....

.....

4 Complete the words using the text on page 41.

- 1 space (paragraph B)
- 2 plant (paragraph B)
- 3 space (paragraph C)

5 Read the short text below and see if your predictions in Activity 3 were correct.

Language Tip

With some compound nouns, the first word can help you understand the second word.

e.g. *space mission*.

There are many different types of missions, but space will tell you the exact type of mission.

Our astronaut Mona is sending us an update on her time in space.



Finally, we have started to see some results from our experiments. Each plant has started to grow, though some are growing much quicker than others.

The corn plant is growing slowly. It appears that it is getting just enough sunlight to grow, but not enough water to grow at a normal rate. I am very disappointed in the results of the corn; I was looking forward to a tasty piece of corn.

The zucchini is receiving enough water, and they are growing. But, because they are not getting enough sunlight, they are growing into very small plants. Zucchini would not be a good food for a long-term space mission. However, they are better than the corn plants.

I am happy to see the potatoes are growing the best. Actually, they are growing at the same rate as they would in nature. I am very happy to see this. It might be a very important breakthrough. And, I think we'll make some into French fries for dinner!

6 Can you find another example of a compound noun in the text?

7 Answer the following questions True (T) or False (F).

- | | |
|--|---|
| 1 Each plant has started to grow at the same speed. | 4 The zucchini plant is not getting enough water, but has enough sunlight. |
| 2 The corn plant is growing quickly. | 5 The zucchini would be a good food for a long-term space mission. |
| 3 The corn plant has enough sunlight to grow, but not enough water. | 6 The potato plants are growing the best. |
| | 7 The corn is growing faster in space than on earth. |
| | 8 Mona will eat corn for dinner. |



2D

What does an astronaut do?

1 Use the text on page 41 to answer the questions below.

1 What is Mona doing at the International Space Station?

.....

2 How fast does a rocket fly?

.....

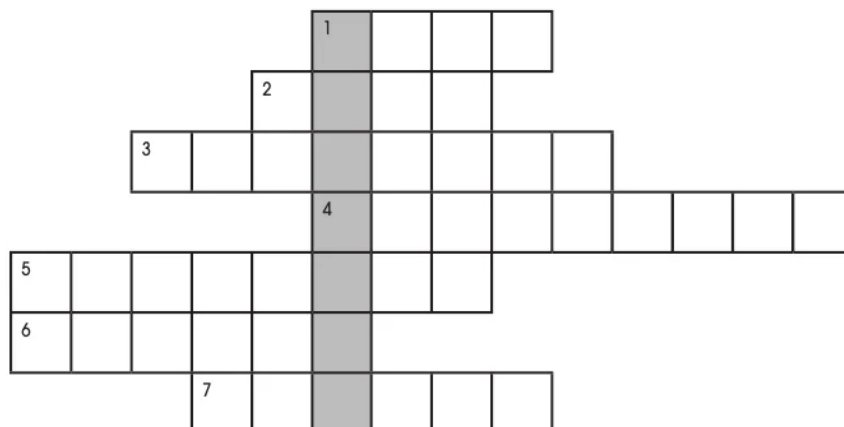
3 Why is Mona doing the experiments?

.....

4 What is Mona growing in the second pot?

.....

2 Write answers using the questions below. In the middle of the crossword you will find a hidden word.



- 1 When plants become bigger
- 2 People on a space mission
- 3 Being in the air unconnected to the ground
- 4 Carrots, peppers, corn
- 5 A type of food that is long and green
- 6 A vehicle that flies to space
- 7 In the air we breathe

3 Write the word found in the middle of the text on the line below and match it to the correct definition.

- is:
- a The force that keeps us connected to the earth
 - b The area between earth and the Moon.
 - c The tail of a rocket.

- 4 Look at the words in the word clouds. Use the text to help you connect the two parts to make a new word.



- 1
- 2
- 3
- 4

- 5
- 6
- 7
- 8

- 5 Write the words from Activity 4 next to their definitions.

- 1 Astronauts do these to research different ideas in space:
.....
- 2 When there is no force that holds you to earth:
.....
- 3 Light that does not come from the sun:
.....
- 4 The area that is beyond the earth's atmosphere:
.....
- 5 Travels astronauts go on to conduct research:
.....

- 6 A place astronauts live in space and conduct research:
.....
- 7 The amount of water and sunlight people needed for things to grow:
.....
- 8 The things plants grow from:
.....

- 6 Read the beginning of an editorial in space. Use your imagination to write a short conclusion to the text. Write between 40-50 words.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Today, we were preparing a new experiment. We wanted to practice what it would be like to enter Mars' atmosphere for a future space mission. We were all sitting down in our stations and were preparing the simulator.



What's life like on Mars?

- What do you know about Mars?
- What do you think life would be like on Mars?
- How can you survive on this planet?

READING

- 1 Read the text and fill in the spaces with the correct details.

It is the year 2055 and planet Earth has become over-populated. The solution to this problem is to find a home on another planet. Mars is the choice. Space travel is the new goal for the UAE and our spaceship, *Explorer One Dubai*, is on its way there now. However, an urgent message was just sent to Mission Control about an engine failure on the spaceship.

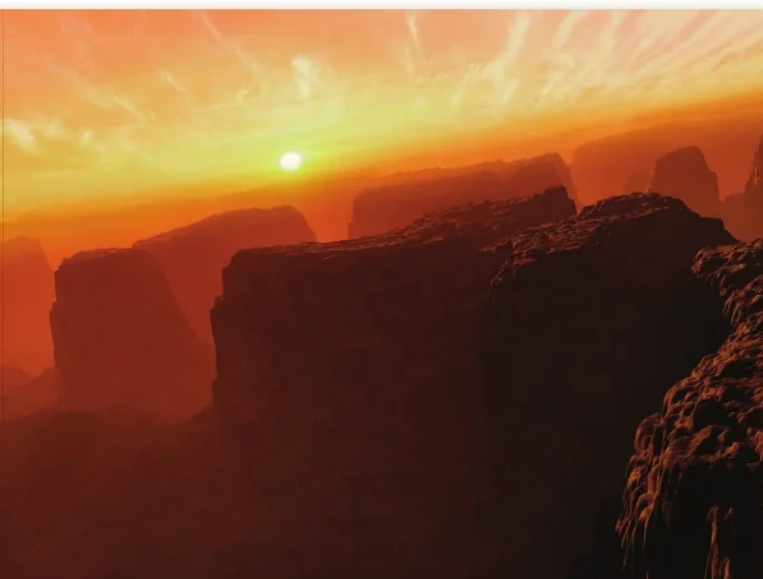
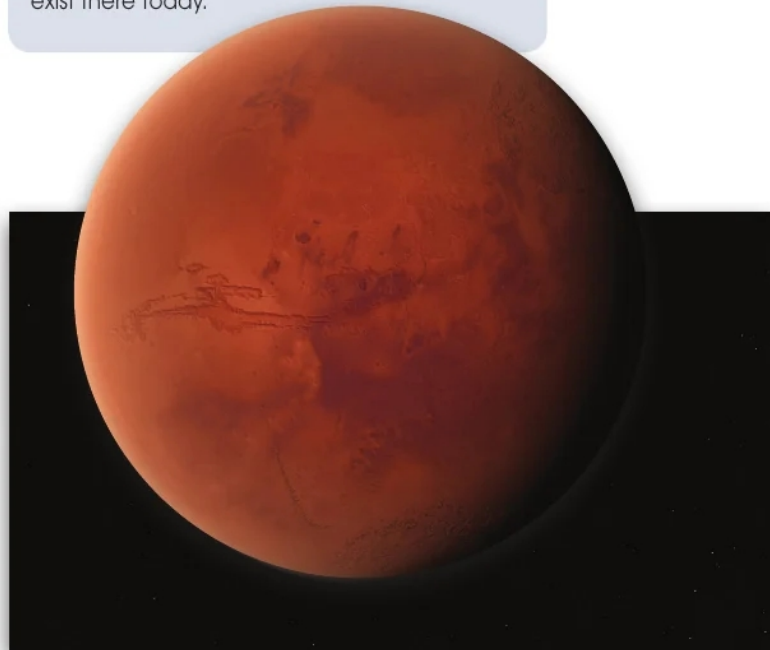
- 1 The year is
- 2 Earth has become
- 3 The solution to this problem is to find a on another
- 4 The name of the spaceship is
- 5 Mission Control received an about an

LISTENING

- 2  Listen to the radio alert from the mission to Mars and answer the questions.

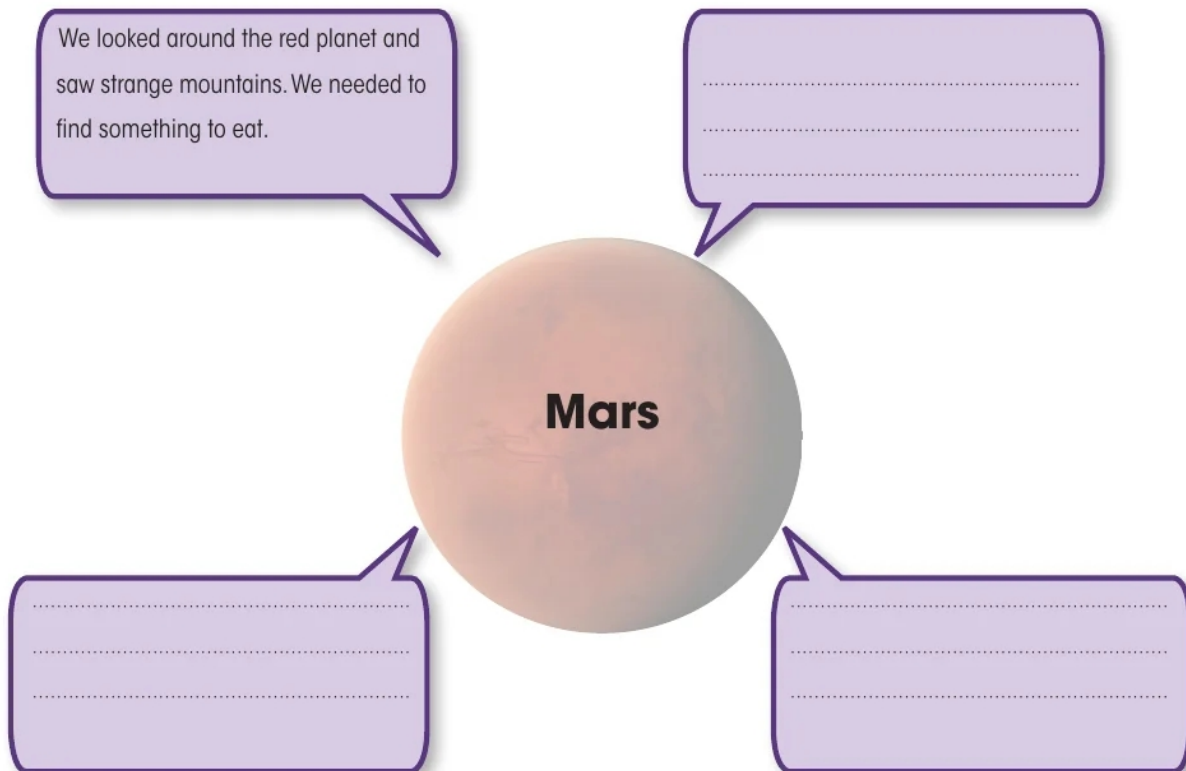
Did You Know?

Some believe that Mars might have had life in the past and may even have living things that exist there today.



- 1 What is the name of the spaceship?
- 2 Where is it from?
- 3 What has happened to the spaceship?
- 4 Is the crew OK? How do we know?
- 5 What will they do now?

- 3  Listen to the urgent message from *Explorer One Dubai* again. Complete a mind map about how you would imagine dealing with the situation. One has been done for you.



WRITING

- 4 Read the beginning of an editorial from the survivors of *Explorer One Dubai*. Use your imagination to write a short conclusion to the text. Write between 40-50 words. Follow your ideas from the mind map and make sure to use expanded noun phrases.

Today, we have decided to do all we can to survive while on this strange red planet. We will build a community while we wait for mission control to send help. We were exploring a bit when suddenly we saw, in the distance, a small jungle and a lake ...

.....

.....

.....

.....

.....

.....

.....

Writing Tip

Remember to use expanded noun phrases in your writing to give the reader more information about what you are describing.



2E

What's life like on Mars?

- 1 Look at each vocabulary word in the box. Fill in the spaces with the correct word.

planets • temperature • climate • gravity • weather • extreme • condition

- 1 The doctor checked thecondition..... of the sick patients on Mars.
- 2 weather on Mars will make it hard for us to survive there.
- 3 Earth, Venus, and Mars are types of
- 4 is the measure of how warm or cold something is.
- 5 Some say change is an important issue on our planet.
- 6 If you don't like the rainy, you shouldn't go out this evening.
- 7 The on both planets is very different. If you weigh 100 lbs on Earth, you would weigh 38 lbs on Mars.

- 2 Look at the example of an informative editorial text about Mars. Underline each noun phrase.

Mars, The Mysterious Red Planet

Mars is a tiny, rocky planet that is very strange for many of us humans on Earth. It is also known as the *Red Planet* because it is covered with rust-like dust. There are no oceans there and the temperature is always very cold. The air is so thin that we couldn't even breathe it. Mars is smaller than the Earth, but larger than our moon. There are extreme weather conditions. It has very strong winds and huge dust storms that can often blow over the whole planet for months! There is so much more to find out about this mysterious planet called Mars.

- 3 Look at the above text. Match the details with the correct noun.

- | | |
|--|--------------|
| 1 small, rocky | a air |
| 2 extreme | b Red Planet |
| 3 covered with rust-like dust | c wind |
| 4 can blow over the whole planet | d Mars |
| 5 so thin ... couldn't even breathe it | e weather |

- 4 Put one line under the nouns, and two lines under the additional details.

- 1 We need to find water on Mars.
- 2 On Mars, we decide to plant our own vegetables.
- 3 We planned to build our houses on Mars.
- 4 On Mars, we need to have clean oxygen.
- 5 Our spaceship crash landed on Mars.
- 6 We are happy to be alive.
- 7 Mars looks very mysterious.
- 8 The strange planet is very red.



5 Expand each noun phrase even more. One has been done for you.

- | | |
|----------------------|--|
| 1 the red planet | a the red planet in space |
| 2 the best astronaut | b |
| 3 a small robot | c |
| 4 the old spaceship | d |
| 5 a beautiful sunset | e |
| 6 many years | f |

6  First, unscramble the words to make a complete sentence. Then, divide the main noun from the additional information.

- 1 up / students / stars / loved / The / look / the / at / to
 a **The students loved to look up at the stars.**
 b
- 2 moon / The / very / bright / is / tonight
 a
 b
- 3 mysterious / very / Mars / is / planet / a
 a
 b
- 4 cold is / weather / very / The / there
 a
 b
- 5 also / Mars / called / is / Red / the / Planet
 a
 b



Review

Space

ENGLISH

VOCABULARY

- 1 Use the words in the word box to complete the sentences.

clip • inches • scientists • research
golden • astronauts • discs

- 1 Look at this video of a rocket launch I found. It is really interesting.
- 2 One meter is 39.37
- 3 The did a lot of before starting the experiment.
- 4 For graduation, my parents gave me a beautiful watch.
- 5 must do a lot of training to prepare for living in space.
- 6 Do you still have your movies and music on ? Or are they digital?

LANGUAGE FOCUS

- 2 Match words from Column A to Column B to create a new word. Write the words on the lines below.

Column A	Column B
space	engine
outer	capsule
artificial	gravity
growing	gravity
rocket	light
zero	space

.....

.....

.....

.....

.....

.....

- 3 Fill in the blanks with words from Activity 2.

- 1 Astronauts travel to in rockets.
- 2 launch rockets from Earth.
- 3 Astronauts return to Earth in a
- 4 are different for corn and potatoes.
- 5 At night I use from a lamp to do my homework.
- 6 In the International Space Station, astronauts live in They float everywhere!

- 4 Match the sentence halves.

- | | |
|---------------------------------|--|
| 1 If I had one million dirhams, | a if you had a one-year vacation? |
| 2 Where would you travel | b I would ask for their autograph. |
| 3 If I met my favorite actor, | c I would take all my friends and family on a nice vacation to the Maldives. |
| 4 I would do better on my exams | d if I studied more. |

- 5 Complete the sentences using your own ideas.

- 1 If I had one million dirhams,
.....
- 2 If I met my favorite actor/actress,
.....
- 3 If I had a year-long vacation,
.....

SPEAKING

6 In pairs, ask and answer the following questions with a partner:

- 1 What planet would you travel to if you had a rocket?
- 2 Who would you take with you on a space mission to Mars?
- 3 What types of food would you take if you went on a space mission?
- 4 What science experiment would you do if you lived on the International Space Station?

WRITING

7 Read the text below. Write a conclusion to the story (50-75 words).

Today, we were preparing a new experiment. We wanted to practice what it would be like to land on Mars. This will help astronauts on future space missions. We were all sitting down in our stations and were preparing the computer testing program. It was going well when suddenly, we heard an alarm ...

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Self-assessment

Read the statements and tick the best description.

I can listen to a radio documentary and predict the order of events.

I can discuss and seek opinions about what to include in a space time capsule.

I can read for specific information that uses the second conditional and speak using the second conditional.

I can read for gist and then make predictions.

I can write an informative text.

Yes Maybe No

☐	☐	☐
☐	☐	☐
☐	☐	☐
☐	☐	☐
☐	☐	☐